Bi-Lipschitz Factor Flows: Regularizing the Latent Geometry of Flow-Matching Decoders

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Abstract

Flow matching in its classical form is a generative model that does not require a latent representation. However, the analysis, interpolation and manipulation of factors in data require one, and its geometry is important. Indeed, in an autoencoder setting, a direction the decoder crushes can be moved without changing the output, while an over-expanded one can turn a small edit into an uncontrolled change after decoding. We ask when a flow-matching decoder, whose output is the endpoint of an ODE, is bi-Lipschitz in its latent code. The two sides of the bi-Lipschitz inequality behave differently. For the expansive bound (i.e. the upper Lipschitz bound) we show that enforcing local conditions on the velocity field suffices. For the contractive bound (i.e. the lower Lipschitz bound), we give a counterexample showing that local control of the velocity field does not prevent collapse at the endpoint. This is how we build BFF (Bi-Lipschitz Factor Flows). On two synthetic benchmarks and on Tabula Muris Senis, BFF reduces empirical decoder distortion by 29–40% with comparable reconstruction and stabilizes the recovery of a sub-dominant factor present in the data and never shown to the model.

1 Introduction

In an autoencoder setting, we can define a Flow Matching [Lipman et al., 2023, Liu et al., 2023, Albergo and Vanden-Eijnden, 2023] based decoder that learns a velocity field transporting a simple source to the data distribution. In this context, for latent factor analysis and manipulation, the decoder's geometry matters and conditions directly the organization of the latent space. This motivates a bi-Lipschitz condition $c\|\mathbf{z}-\mathbf{z}'\| \le \|D(\mathbf{z})-D(\mathbf{z}')\| \le C\|\mathbf{z}-\mathbf{z}'\|$. For an explicit decoder, its two sides can be controlled through the extreme singular values of its Jacobian [Rifai et al., 2011, Miyato et al., 2018, Behrmann et al., 2019, Zhan et al., 2026]. However for a flow decoder, the Jacobian is no longer in closed form as it is the terminal value of a sensitivity accumulated all along an ODE trajectory. We ask which local properties of the velocity field certify the endpoint geometry. We then show that pointwise bounds control expansion but cannot prevent contractive collapse. This gives us BFF (§4), whose regularizers target the resulting local and endpoint conditions.

2 Setup: the decoder as an ODE endpoint

An encoder $E_\theta : \mathbb{R}^N \to \mathbb{R}^d$, $d \ll N$, maps a target observation $\mathbf{x}_1$ to a code $\mathbf{z} = E_\theta(\mathbf{x}_1)$. This code then conditions a velocity field since the decoder is the endpoint of the following ODE:

$$\frac{d\mathbf{X}_t}{dt} = \mathbf{v}_\theta(t, \mathbf{X}_t, \mathbf{z}), \qquad \mathbf{X}_0 = \mathbf{x}_0, \qquad t \in [0,1], \qquad \widehat{\mathbf{x}} = \Phi_{\mathbf{z}}(\mathbf{x}_0) := \mathbf{X}_1. \tag{1}$$

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Throughout our study, the source $\mathbf{x}_0$ is fixed. This isolates the dependence on $\mathbf{z}$ from the variability of the source. Then, for this fixed $\mathbf{x}_0$, $F : \mathbf{z} \mapsto \Phi_{\mathbf{z}}(\mathbf{x}_0)$ plays exactly the role of the decoder D_ϕ from the deterministic autoencoder setting. Its Jacobian is the terminal value of the sensitivity $\mathbf{S}_t = \partial_{\mathbf{z}} \mathbf{X}_t \in \mathbb{R}^{N \times d}$, which measures how the state $\mathbf{X}_t$ varies with respect to the latent code $\mathbf{z}$. Differentiating (1) in $\mathbf{z}$ using the chain rule and that $\partial_{\mathbf{z}} \mathbf{X}_0 = 0$, we obtain:

$$\dot{\mathbf{S}}_t = \underbrace{\partial_{\mathbf{x}} \mathbf{v}_\theta(t, \mathbf{X}_t, \mathbf{z})}_{=\mathbf{A}(t) \in \mathbb{R}^{N \times N}} \mathbf{S}_t \; + \; \underbrace{\partial_{\mathbf{z}} \mathbf{v}_\theta(t, \mathbf{X}_t, \mathbf{z})}_{=\mathbf{B}(t) \in \mathbb{R}^{N \times d}}, \qquad \mathbf{S}_0 = \mathbf{0}. \tag{2}$$

And then we obtain from the variation of constants formula and denoting $\Psi(t, s)$ the transition matrix of the homogeneous part of the linear differential equation (2):

$$\mathbf{S}_t = \int_0^t \Psi(t, s)\, \mathbf{B}(s)\, ds \tag{3}$$

The integral of equation 3 shows the difficulty: $\mathbf{S}_1$ is an accumulation of instantaneous latent sensitivities $\mathbf{B}(s)$ over time starting at $t = 0$ and each transported to $t = 1$. The bi-Lipschitz statement therefore becomes a statement about this integral and the two sides of the inequality interact with it in different ways.

3 Local conditions cannot certify the contractive side

The expansive side is guaranteed pointwise. Norms pass through (3) by the triangle inequality and Grönwall's lemma in its differential form bounds the transition matrix. We obtain the inequality: $\|\Psi(t, s)\|_2 \leq \exp(\int_s^t \|\mathbf{A}(\tau)\|_2 d\tau)$.

Proposition 1 (Upper bound)**.** *If $\|\mathbf{A}(t)\|_2 \leq C_x$ and $\|\mathbf{B}(t)\|_2 \leq C_z$ for t in $[0, 1]$, then we have $\sigma_{\max}(\mathbf{S}_t) \leq \frac{C_z}{C_x}(e^{C_x t} - 1) \leq C_z\, t\, e^{C_x t}$ for $C_x > 0$, and $\sigma_{\max}(\mathbf{S}_t) \leq C_z t$ for $C_x = 0$.*

Hence, two pointwise and cheaply estimated quantities control the expansion of the fully integrated decoder (Full proof in App. B).

The contractive side can only be similarly guaranteed near $t = 0$, the beginning of the trajectory. Since $\mathbf{S}_0 = 0$, we have $\dot{\mathbf{S}}_0 = \mathbf{B}(0)$ and, near $t = 0$ under mild regularity, $\mathbf{S}_t = t\, \mathbf{B}(0) + \mathbf{R}(t)$ with $\|\mathbf{R}(t)\|_2 = o(t)$.

Proposition 2 (Local lower bound)**.** *Assume $\mathbf{A}$ is locally bounded near $t = 0$, $\mathbf{B}$ is continuous at $t = 0$ along the trajectory, and $\sigma_{\min}(\mathbf{B}(0)) \geq c_z > 0$. Then there is $t^\star > 0$ with $\sigma_{\min}(\mathbf{S}_t) \geq \frac{c_z}{2}\, t$ for all $t \in [0, t^\star]$.*

However, this bound is strictly local and nothing guarantees it will hold along the trajectory i.e. until $t = 1$ (App. C).

It cannot be globalized. Contributions $\Psi(1, s)\mathbf{B}(s)$ can cancel and this is a property of the formulation. We exhibit a counterexample below.

Proposition 3 (No pointwise lower bound certifies $\sigma_{\min}(\mathbf{S}_1) > 0$)**.** *There is a smooth, bounded in $\mathbf{x}$ velocity field with $\sigma_{\min}(\partial_{\mathbf{z}} \mathbf{v}_\theta(t, \cdot, \cdot)) = 1$ for all $t \in [0, 1]$ whose flow map satisfies $\mathbf{S}_1 = \mathbf{0}$ for every $\mathbf{z}$, i.e. for which $\Phi_{\mathbf{z}}(\mathbf{x}_0)$ is constant in $\mathbf{z}$.*

Construction. Take $d = 2$, any $N \geq 2$, $\mathbf{P} \in \mathbb{R}^{N \times 2}$ with orthonormal columns, and $\mathbf{v}_\theta(t, \mathbf{x}, \mathbf{z}) = \mathbf{P}\, \mathbf{R}_{2\pi t}\, \mathbf{z}$ with $\mathbf{R}_\alpha \in \mathbb{R}^{2 \times 2}$ the rotation matrix of angle α and $\mathbf{z} \in \mathbb{R}^d = \mathbb{R}^2$ the latent code. Then $\mathbf{A}(t) = \partial_{\mathbf{x}} \mathbf{v}_\theta = 0$, so $\Psi(1, s) = \mathbf{I}$, and $\mathbf{B}(t) = \mathbf{P}\mathbf{R}_{2\pi t}$ has both singular values equal to 1 for every t. Yet $\mathbf{S}_1 = \mathbf{P} \int_0^1 \mathbf{R}_{2\pi s} ds = \mathbf{0}$ and $\Phi_{\mathbf{z}}(\mathbf{x}_0) = \mathbf{x}_0$ for every $\mathbf{z}$. □

We see that every intermediate state genuinely depends on $\mathbf{z}$ and the dependence is undone exactly at the endpoint.

What has to be controlled instead. The cancellation above happens since $\mathbf{S}_1$ is an average of the transported contributions $\mathbf{M}(s) = \Psi(1,s)\mathbf{B}(s)$ and local conditions can only control how large each contribution is. The two are separated by a variance: for every unit $\mathbf{v}$,

$$\underbrace{\int_0^1 \|\mathbf{M}(s)\mathbf{v}\|_2^2\, ds}_{\text{size of the contributions}} = \underbrace{\|\mathbf{S}_1\mathbf{v}\|_2^2}_{\text{what reaches the endpoint}} + \underbrace{\int_0^1 \|\mathbf{M}(s)\mathbf{v} - \mathbf{S}_1\mathbf{v}\|_2^2\, ds}_{\text{spread around the average}}. \tag{4}$$

Keeping the contributions large is therefore useless if their spread absorbs all of it and this is what happens in Prop. 3 where, for every $\mathbf{v}$ the left-hand side equals 1 and the two right-hand terms equal 0 and 1. We see that we need to control the square of the average and that this cannot be done by a pointwise condition on the velocity field (cf App. E). Therefore, controlling this squared average can only be done after integration as the spread term would require the sensitivities $\mathbf{M}(s)$ along the whole trajectory. That is what the endpoint term $\mathcal{L}_{\text{sep}}$ introduced in §4 is designed for.

4 Bi-Lipschitz Factor Flows

BFF pairs a deterministic encoder with the flow decoder (1) and adds four geometric terms (Fig. 2). Terms (i)–(ii) target the quantities associated with C_x and C_z assumed in Prop. 1, term (iii) enforces c_z near $t = 0$ as required by Prop. 2 and term (iv) acts after integration by penalizing the endpoint collapse exposed by Prop. 3.

(i) A surrogate for C_x: $\mathcal{L}_{Jx} = \mathbb{E}[\|\partial_x v_\theta\|_F^2]$, a computable surrogate through the Hutchinson Estimator [Hutchinson, 1989] since $\|J\|_2 \le \|J\|_F$ (this is inspired by the Jacobian penalty of Finlay et al. [2020]).

(ii) Latent Jacobian, upper, controlling C_z: $\mathcal{L}^u_{J_z} = \mathbb{E}[\,\mathrm{ReLU}(\sigma_{\max}(\partial_{\mathbf{z}}\mathbf{v}_\theta) - C_z)^2]$.

(iii) Latent Jacobian, early, the regime of Prop. 2: $\mathcal{L}^\ell_{J_z,\text{early}} = \mathbb{E}_{t\sim\mathcal{U}([0,\varepsilon])}\mathrm{ReLU}(c_z - \sigma_{\min}(\partial_{\mathbf{z}}\mathbf{v}_\theta))^2$.

(iv) Endpoint separation (cf Prop. 3) over within-batch pairs $\mathcal{P}_K$ decoded from the same source as a sampled relaxation of the lower bi-Lipschitz inequality on the integrated decoder (on the absolute scale m_z, see App. H)

$$\mathcal{L}_{\text{sep}} = \frac{1}{|\mathcal{P}_K|}\sum_{(i,j)\in\mathcal{P}_K} \mathrm{ReLU}\big(m_z\|\mathbf{z}_i - \mathbf{z}_j\|_2 - \|\widehat{\mathbf{x}}_i - \widehat{\mathbf{x}}_j\|_2\big)^2\, \mathbf{1}_{\{\|\mathbf{z}_i-\mathbf{z}_j\|_2>\delta_z\}} \tag{5}$$

The objective is $\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda_{\text{FM}}\mathcal{L}_{\text{FM}} + \lambda_x\mathcal{L}_{J_x} + \lambda^u_z\mathcal{L}^u_{J_z} + \lambda^\ell_z\mathcal{L}^\ell_{J_z,\text{early}} + \lambda_{\text{sep}}\mathcal{L}_{\text{sep}}$

with $\mathcal{L}_{\text{rec}} = \mathbb{E}\|\mathbf{x}_1 - \Phi_{E_\theta(\mathbf{x}_1)}(\mathbf{x}_0)\|^2$ and $\mathcal{L}_{\text{FM}} = \mathbb{E}\|\mathbf{v}_\theta(t,\mathbf{x}_t,\mathbf{z}) - (\mathbf{x}_1 - \mathbf{x}_0)\|^2$ on linear paths $\mathbf{x}_t = (1-t)\mathbf{x}_0 + t\mathbf{x}_1$.

5 Experiments

We compare an unregularized model trained $\mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{FM}}$ only (**Rec + FM**) against the fully regularized model (**BFF**). The metrics we report are: MSE, the R^2 of a linear probe predicting a held-out factor never given to the model from $\mathbf{z}$, and an empirical distortion ratio κ^{95}_{emp} (with $r_{ij} = \|\widehat{\mathbf{x}}_i - \widehat{\mathbf{x}}_j\|_2/\|\mathbf{z}_i - \mathbf{z}_j\|_2$ over test pairs, and $\kappa^{95}_{\text{emp}} = r_{\text{p95}}/r_{\text{p05}}$ measures how unevenly the decoder expands directions, $\kappa^{95}_{\text{emp}} \approx 1$ being the best, cf App. G).

Table 1: All three benchmarks (mean ± s.d.; 3 seeds for genes, 5 otherwise).

	Configuration	MSE (↓)	$\mathbf{R^2_a}$ (↑)	$\kappa^{\mathbf{95}}_{\text{emp}}$ (↓)
genes	Rec + FM	0.0253 ± 0.0017	0.485 ± 0.350	3.67 ± 1.07
	BFF	$\mathbf{0.0238 \pm 0.0002}$	$\mathbf{0.982 \pm 0.001}$	$\mathbf{2.20 \pm 0.01}$
	ablation: no $\mathcal{L}_{\text{sep}}$	0.0256 ± 0.0003	0.340 ± 0.131	2.51 ± 0.08
img.	Rec + FM	$(1.2 \pm 0.5)\cdot 10^{-5}$	0.472 ± 0.220	1.95 ± 0.13
	BFF	$\mathbf{(0.8 \pm 0.2)\cdot 10^{-5}}$	$\mathbf{0.705 \pm 0.031}$	$\mathbf{1.38 \pm 0.06}$
TMS	Rec + FM	86.06 ± 4.28	—	3.123 ± 0.058
	BFF	$\mathbf{84.40 \pm 3.70}$	—	$\mathbf{2.176 \pm 0.038}$

Synthetic benchmarks with known factors. Two factors $(t, a) \sim \mathcal{U}([0,1]^2)$ generate both synthetic transcriptomes and rendered images where a is deliberately sub-dominant (App. G.3). A pixel-level probe gives $R_t^2 = 1.000$ and $R_a^2 = 0.847$, which confirms that a is present in the observations. In Table 1 reconstruction is comparable while κ_{emp}^{95} drops by 40% and 29% respectively. We also observe a change in the stability of the representation: without the bi-Lipschitz regularization, the recovery of the subdominant factor a depends heavily on seed initialization ($R_a^2 = 0.472 \pm 0.220$ on images) whereas BFF lowers the spread by a factor 7 (0.705 ± 0.031). This effect is visually clear in Fig. 3 (App. J.1) where we see how BFF turns a compressed and disordered a-direction into a more spread-out latent axis with a monotone gradient.

The ablation row on the synthetic genes data was made to test the effectiveness of the lower-bound loss construction. Removing $\mathcal{L}_{\text{sep}}$ still improves conditioning (κ_{emp}^{95} : $3.67 \rightarrow 2.51$) but fails to recover a ($R_a^2 = 0.340$, which does not improve the unregularized baseline). We see how local terms can bound the integrand and control the expansion but can still leave the endpoint free to cancel. Therefore, we observe experimentally that local conditioning alone does not prevent a latent direction from being crushed.

Real transcriptomes: Tabula Muris Senis. On TMS, we train on $96{,}768$ cells from 11 tissues. Their age is never given to the model. Age here is treated as a potentially subdominant factor of variation in the data. The evaluation is performed on held-out animals over 29 tissue $\times$ cell-type contexts (App. G.4). Reconstruction is comparable while regularization lowers κ_{emp}^{95} by 30% (Table 1). The unregularized model is very close to the input-space age geometry while BFF improves all three age-axis diagnostics (Table 3) and yields a more ordered progression of chronological ages along the fitted age direction (Fig. 1).

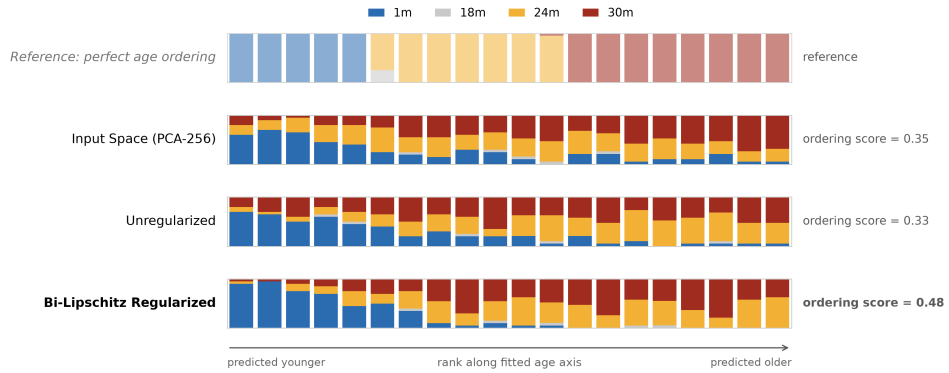


Figure 1: Tabula Muris Senis, one tissue $\times$ cell-type context. Cells are ranked along an age direction fitted on training animals and grouped into bins from predicted younger to predicted older. Colors show the age composition of each bin. The top row is the perfect-ordering reference. BFF yields a more ordered age progression and tends to reduce the mixing between the oldest and youngest cells. The ordering score reported equals $2\text{AUC} - 1$ over all pairs of age groups (0 corresponds to random and 1 to perfect ordering).

6 Discussion

In an autoencoder setting, many latent organizations reconstruct equally well, and some of them represent poorly the factors that contribute little to the reconstruction error. Therefore, introducing geometric regularization allows us to select better-conditioned ones. However, there are two limits to our claim. Firstly, geometric regularity does not mean interpretability [Locatello et al., 2019], so on real data we report better organization relative to an external read-out but not identification. Then, our lower-side regularization remains partial: Prop. 2 is local and $\mathcal{L}_{\text{sep}}$ is a sampled endpoint surrogate. Prop. 4 (App. E) isolates what is missing: keeping the transported sensitivities large does not prevent them from cancelling at the endpoint. The next steps could be to penalize an estimator of the spread term in (4) directly (which would cost additional sensitivity computations) or to design flows with an architecture that would prevent cancellation by construction. This could then provide a genuine global bi-Lipschitz guarantee.

References

Michael S. Albergo and Eric Vanden-Eijnden. Building normalizing flows with stochastic interpolants. In *International Conference on Learning Representations (ICLR)*, 2023.

Georgios Arvanitidis, Lars Kai Hansen, and Søren Hauberg. Latent space oddity: on the curvature of deep generative models. In *International Conference on Learning Representations (ICLR)*, 2018.

Jens Behrmann, Will Grathwohl, Ricky T. Q. Chen, David Duvenaud, and Jörn-Henrik Jacobsen. Invertible residual networks. In *International Conference on Machine Learning (ICML)*, 2019.

Peter D. Caie, Rebecca E. Walls, Alexandra Ingleston-Orme, Sandeep Daya, Tom Houslay, Rob Eagle, Mark E. Roberts, and Neil O. Carragher. High-content phenotypic profiling of drug response signatures across distinct cancer cells. *Molecular Cancer Therapeutics*, 9(6):1913–1926, 2010.

Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David Duvenaud. Neural ordinary differential equations. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

Chris Finlay, Jörn-Henrik Jacobsen, Levon Nurbekyan, and Adam M. Oberman. How to train your neural ODE: the world of Jacobian and kinetic regularization. In *International Conference on Machine Learning (ICML)*, 2020.

Geoffrey E. Hinton and Ruslan R. Salakhutdinov. Reducing the dimensionality of data with neural networks. *Science*, 313(5786):504–507, 2006.

Michael F. Hutchinson. A stochastic estimator of the trace of the influence matrix for Laplacian smoothing splines. *Communications in Statistics – Simulation and Computation*, 18(3):1059–1076, 1989.

Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR)*, 2023.

Xingchao Liu, Chengyue Gong, and Qiang Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *International Conference on Learning Representations (ICLR)*, 2023.

Francesco Locatello, Stefan Bauer, Mario Lucic, Gunnar Rätsch, Sylvain Gelly, Bernhard Schölkopf, and Olivier Bachem. Challenging common assumptions in the unsupervised learning of disentangled representations. In *International Conference on Machine Learning (ICML)*, 2019.

Romain Lopez, Jeffrey Regier, Michael B. Cole, Michael I. Jordan, and Nir Yosef. Deep generative modeling for single-cell transcriptomics. *Nature Methods*, 15:1053–1058, 2018.

Mohammad Lotfollahi, F. Alexander Wolf, and Fabian J. Theis. scGen predicts single-cell perturbation responses. *Nature Methods*, 16:715–721, 2019.

Takeru Miyato, Toshiki Kataoka, Masanori Koyama, and Yuichi Yoshida. Spectral normalization for generative adversarial networks. In *International Conference on Learning Representations (ICLR)*, 2018.

Konpat Preechakul, Nattanat Chatthee, Suttisak Wizadwongsa, and Supasorn Suwajanakorn. Diffusion autoencoders: Toward a meaningful and decodable representation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10619–10629, 2022.

Salah Rifai, Pascal Vincent, Xavier Muller, Xavier Glorot, and Yoshua Bengio. Contractive auto-encoders: Explicit invariance during feature extraction. In *International Conference on Machine Learning (ICML)*, 2011.

Hang Shao, Abhishek Kumar, and P. Thomas Fletcher. The Riemannian geometry of deep generative models. In *IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2018.

Eduardo D. Sontag. *Mathematical Control Theory: Deterministic Finite Dimensional Systems*. Springer, New York, 2nd edition, 1998.

Rashmi Sreeramachandra Murthy, Shobana V. Stassen, Dickson M. D. Siu, Michelle C. K. Lo, Gwinky G. K. Yip, and Kevin K. Tsia. Generalizable morphological profiling of cells by interpretable unsupervised learning. *Nature Communications*, 16:11465, 2025.

The Tabula Muris Consortium. A single-cell transcriptomic atlas characterizes ageing tissues in the mouse. *Nature*, 583:590–595, 2020.

Qipeng Zhan, Zhuoping Zhou, Zexuan Wang, Qi Long, and Li Shen. Bi-Lipschitz autoencoder with injectivity guarantee. In *International Conference on Learning Representations (ICLR)*, 2026.

Yuhui Zhang, Yuchang Su, Chenyu Wang, Tianhong Li, Zoe Wefers, Jeffrey J. Nirschl, James Burgess, Daisy Ding, Alejandro Lozano, Emma Lundberg, and Serena Yeung-Levy. CellFlux: Simulating cellular morphology changes via flow matching. In *International Conference on Machine Learning (ICML)*, 2025.

Appendix

A Related work

Latent geometry of autoencoders. A plain autoencoder [Hinton and Salakhutdinov, 2006] minimizing $\mathbb{E}[\|\mathbf{x} - D(E(\mathbf{x}))\|^2]$ does not constrain the geometry of the latent encoding. Several papers work on regularizing the Jacobian or the spectrum of the maps involved: contractive autoencoders [Rifai et al., 2011], spectral normalization [Miyato et al., 2018] and invertible residual networks [Behrmann et al., 2019]. The work which is the closest to our BFF and from which we directly inspired ourselves is BLAE [Zhan et al., 2026]. BLAE combines an encoder-side separation constraint with an explicit bi-Lipschitz penalty on the decoder, $\mathbb{E}_{\mathbf{z}}[\mathrm{ReLU}(\kappa^{-1} - \sigma_{\min}(J_{D_\phi}))^2 + \mathrm{ReLU}(\sigma_{\max}(J_{D_\phi}) - \kappa)^2]$. Their choice of regularizing the decoder rather than the encoder is one that we made too for structural reasons. Indeed, a continuous map $\mathbb{R}^m \to \mathbb{R}^n$ with $n < m$ cannot be injective on the whole ambient space so it cannot be bi-Lipschitz there. However, on a data manifold of lower dimension, the encoder can be injective, and if its restriction to it is bi-Lipschitz then its inverse is. Therefore, controlling the decoder controls both directions on the learned manifold. What does not transfer from BLAE to our setting is the regularization principle: they have a network Jacobian at their disposal (J_{D_ϕ}) whereas the Jacobian of our decoder is the terminal value of the sensitivity (3).

The latent space as a Riemannian manifold. A related line of work studies latent geometry explicitly. It equips the latent space with the decoder-induced metric $J_D^\top J_D$ and computes distances and interpolations in this geometry rather than with the Euclidean metric [Arvanitidis et al., 2018, Shao et al., 2018]. This is complementary to our approach: these methods account for the learned geometry at analysis time, whereas bi-Lipschitz regularization aims to make Euclidean latent geometry itself better conditioned. Indeed, $c^2 I \preceq J_D^\top J_D \preceq C^2 I$ means that the induced metric remains within a bounded distortion of the Euclidean one.

For a flow decoder, the induced metric $\mathbf{S}_1^\top \mathbf{S}_1$ is not available in closed form. Our local penalties instead have an impact on the time-averaged quantity $\mathbf{G} = \int_0^1 \mathbf{M}(s)^\top \mathbf{M}(s)\, ds$. Prop. 4 shows that $\mathbf{S}_1^\top \mathbf{S}_1 \preceq \mathbf{G}$, with the gap given by the spread term in (4) (App. E).

Learned codes conditioning an iterative decoder. The closest architecture to ours outside flow matching is the diffusion autoencoder of Preechakul et al. [2022], which combines a learned semantic code with a diffusion-based decoder. With our fixed source $\mathbf{x}_0$, decoding similarly depends only on the learned code, so $F : \mathbf{z} \mapsto \Phi_{\mathbf{z}}(\mathbf{x}_0)$ defines a decoder as in §2.

Flow matching and conditioning. Flow matching [Lipman et al., 2023] and its relatives [Liu et al., 2023, Albergo and Vanden-Eijnden, 2023] learn a velocity field transporting a simple source onto the data distribution, within the more general family of ODE-defined neural models [Chen et al., 2018]. Conditioning the velocity field on an external variable is done by CellFlux [Zhang et al., 2025] as they condition on a perturbation embedding for cellular imaging for example. Our aim is to replace the externally supplied condition by a code $\mathbf{z} = E_\theta(\mathbf{x}_1)$ learned from the data, and then to ask what its geometry looks like. In the single-cell literature, latent-variable generative models such as scVI [Lopez et al., 2018] and scGen [Lotfollahi et al., 2019] learn representations but with explicit decoders; MorphoGenie [Sreeramachandra Murthy et al., 2025] builds interpretable morphological latents with a GAN-based generator.

B Proof of Proposition 1

Fix $s \leq t$ and a unit vector $\mathbf{y}_0 \in \mathbb{R}^N$, and let $\mathbf{y}(\tau) = \Psi(\tau, s)\mathbf{y}_0$. We then have $\dot{\mathbf{y}} = \mathbf{A}(\tau)\mathbf{y}$, $\mathbf{y}(s) = \mathbf{y}_0$. We derive

$$\frac{d}{d\tau}|\mathbf{y}(\tau)|_2^2 = 2\langle \mathbf{y}(\tau), \mathbf{A}(\tau)\mathbf{y}(\tau)\rangle \leq 2\|\mathbf{y}(\tau)\|_2 \|\mathbf{A}(\tau)\mathbf{y}(\tau)\|_2 \leq 2\|\mathbf{A}(\tau)\|_2 \|\mathbf{y}(\tau)\|_2^2, \tag{6}$$

using bilinearity of the scalar product, Cauchy–Schwarz inequality and the sub-multiplicativity of the spectral norm. We then apply Grönwall's lemma in differential form (if $u'(\tau) \leq a(\tau)u(\tau)$, then

$u(t) \le u(s)\exp(\int_s^t a(\tau)d\tau)$) to the scalar quantity $\|\mathbf{y}(\tau)\|_2^2$ and this gives

$$\|\mathbf{y}(t)\|_2^2 \le \|\mathbf{y}(s)\|_2^2 \exp(2\int_s^t \|\mathbf{A}(\tau)\|_2 d\tau) \tag{7}$$

Now, by applying the square root function and taking the supremum over unit $\mathbf{y}_0$,

$$\|\Psi(t,s)\|_2 \;\le\; \exp\Big(\int_s^t \|\mathbf{A}(\tau)\|_2\, d\tau\Big). \tag{8}$$

We then apply the integral triangle inequality and sub-multiplicativity to (3),

$$\|\mathbf{S}_t\|_2 \le \int_0^t \|\Psi(t,s)\|_2 \|\mathbf{B}(s)\|_2\, ds \le \int_0^t \exp\Big(\int_s^t \|\mathbf{A}(\tau)\|_2 d\tau\Big)\|\mathbf{B}(s)\|_2\, ds. \tag{9}$$

If $\|\mathbf{A}\|_2 \le C_x$ and $\|\mathbf{B}\|_2 \le C_z$ this gives us $\|\mathbf{S}_t\|_2 \le C_z \int_0^t e^{C_x(t-s)} ds = \frac{C_z}{C_x}(e^{C_x t} - 1)$ for $C_x > 0$, and $\|\mathbf{S}_t\|_2 \le C_z t$ for $C_x = 0$ (the simplified form comes from $e^u - 1 \le u\, e^u$ with $u = C_x t$). □

C Proof of Proposition 2

Since $\mathbf{S}_t$ is continuous with $\mathbf{S}_0 = 0$ and $\mathbf{A}$ is locally bounded near $t = 0$, $\mathbf{A}(t)\mathbf{S}_t \to 0$. Also, by continuity of $\mathbf{B}$ at 0, $\mathbf{B}(t) \to \mathbf{B}(0)$. Hence $\dot{\mathbf{S}}_t = \mathbf{A}(t)\mathbf{S}_t + \mathbf{B}(t) \to \mathbf{B}(0)$ as $t \to 0$. Writing

$$\mathbf{S}_t = \int_0^t \dot{\mathbf{S}}_s\, ds = t\,\mathbf{B}(0) + \mathbf{R}(t), \qquad \mathbf{R}(t) = \int_0^t \big(\dot{\mathbf{S}}_s - \mathbf{B}(0)\big) ds, \tag{10}$$

we get $\|\mathbf{R}(t)\|_2/t \le \sup_{0\le s\le t} \|\dot{\mathbf{S}}_s - \mathbf{B}(0)\|_2 \to 0$, i.e. $\|\mathbf{R}(t)\|_2 = o(t)$.

For any unit $\mathbf{v} \in \mathbb{R}^d$, the reverse triangle inequality ($\|a+b\| \ge \big|\|a\| - \|b\|\big| \ge \|a\| - \|b\|$) gives

$$\|\mathbf{S}_t\mathbf{v}\|_2 \ge t\|\mathbf{B}(0)\mathbf{v}\|_2 - \|\mathbf{R}(t)\mathbf{v}\|_2 \ge t\,\sigma_{\min}(\mathbf{B}(0)) - \|\mathbf{R}(t)\|_2, \tag{11}$$

and taking the infimum over unit $\mathbf{v}$, $\sigma_{\min}(\mathbf{S}_t) \ge t\,\sigma_{\min}(\mathbf{B}(0)) - \|\mathbf{R}(t)\|_2$. If $\sigma_{\min}(\mathbf{B}(0)) \ge c_z > 0$, then since $\|\mathbf{R}(t)\|_2 = o(t)$ there is $t^\star > 0$ with $\|\mathbf{R}(t)\|_2 \le \frac{c_z}{2} t$ on $[0, t^\star]$, and $\sigma_{\min}(\mathbf{S}_t) \ge \frac{c_z}{2} t$ there. □

D Details on Proposition 3

Details on the counterexample. The field $\mathbf{v}_\theta(t, \mathbf{x}, \mathbf{z}) = \mathbf{P}\mathbf{R}_{2\pi t}\mathbf{z}$ is analytic in t, linear in $\mathbf{z}$, constant in $\mathbf{x}$. Also, the field satisfies $\|\mathbf{v}_\theta(t, \mathbf{x}, \mathbf{z})\| = \|\mathbf{z}\|$ for all $\mathbf{z}$. Both of its Jacobians are exactly the objects the local penalties of §4 see: $\partial_{\mathbf{x}}\mathbf{v}_\theta \equiv 0$ (so $\mathcal{L}_{J_x} = 0$, its optimum) and $\partial_{\mathbf{z}}\mathbf{v}_\theta = \mathbf{P}\mathbf{R}_{2\pi t}$ with $\sigma_{\min} = \sigma_{\max} = 1$ (so $\mathcal{L}^u_{J_z} = \mathcal{L}^\ell_{J_z,\text{early}} = 0$ for $C_z \ge 1 \ge c_z$). All the local penalties are therefore minimized even though the decoder is constant in $\mathbf{z}$. This is why we introduce a loss, $\mathcal{L}_{\text{sep}}$, evaluated after ODE integration. $\mathcal{L}_{\text{sep}}$ remains sensitive to cancellations that the local penalties do not take into account.

Intermediate states do carry latent information. Although the endpoint is independent of $\mathbf{z}$, the latent dependence is active throughout the trajectory and only cancels at $t = 1$. Indeed, since $\mathbf{A}(t) = 0$, we have $\Psi(t, s) = \mathbf{I}$, and therefore

$$\mathbf{S}_t = \int_0^t \mathbf{B}(s) ds = \mathbf{P}\int_0^t \mathbf{R}_{2\pi s} ds. \tag{12}$$

As $\mathbf{R}_{2\pi s} = \left(\begin{smallmatrix} \cos(2\pi s) & -\sin(2\pi s) \\ \sin(2\pi s) & \cos(2\pi s) \end{smallmatrix}\right)$ and integrating each entry from 0 to t, we obtain

$$\mathbf{S}_t = \frac{1}{2\pi}\mathbf{P}\left(\begin{smallmatrix} \sin(2\pi t) & \cos(2\pi t)-1 \\ 1-\cos(2\pi t) & \sin(2\pi t) \end{smallmatrix}\right). \tag{13}$$

For $t \in (0, 1)$, this matrix is nonzero, so the intermediate state $\mathbf{X}_t$ depends on $\mathbf{z}$ and it is only at the end of the trajectory that $\mathbf{S}_1 = \mathbf{0}$. Thus, the latent dependence is active throughout the trajectory but cancels at the endpoint.

E The cancellation gap

We state here the two results summarized in §3. We have $\mathbf{M}(s) = \Psi(1,s)\mathbf{B}(s) \in \mathbb{R}^{N\times d}$, so that $\mathbf{S}_1 = \int_0^1 \mathbf{M}(s)\,ds$, and

$$\mathbf{G} = \int_0^1 \mathbf{M}(s)^\top \mathbf{M}(s)\,ds, \qquad \mathcal{V} = \sup_{\|\mathbf{v}\|_2=1} \int_0^1 \|\mathbf{M}(s)\mathbf{v} - \mathbf{S}_1\mathbf{v}\|_2^2\,ds. \tag{14}$$

We note that $\mathbf{v}^\top\mathbf{G}\mathbf{v}$ measures the total squared transported sensitivity along the latent direction $\mathbf{v}$. This construction is analogous to finite-horizon Gramians in linear systems theory [Sontag, 1998]. And $\mathcal{V}$ measures the temporal spread of the transported sensitivities around their endpoint average in the worst latent direction.

Proposition 4 (Cancellation gap)**.** *For every unit* $\mathbf{v} \in \mathbb{R}^d$*,*

$$\int_0^1 \|\mathbf{M}(s)\mathbf{v}\|_2^2\,ds = \|\mathbf{S}_1\mathbf{v}\|_2^2 + \int_0^1 \|\mathbf{M}(s)\mathbf{v} - \mathbf{S}_1\mathbf{v}\|_2^2\,ds.$$

Consequently,

$$\sigma_{\min}(\mathbf{S}_1)^2 \geq \lambda_{\min}(\mathbf{G}) - \mathcal{V}. \tag{15}$$

Corollary 1. *Assume* $\|\mathbf{A}(t)\|_2 \leq C_x$ *on* $[0,1]$.

(a) If $\sigma_{\min}(\mathbf{B}(t)) \geq c_z$ *on all of* $[0,1]$*, then* $\lambda_{\min}(\mathbf{G}) \geq c_z^2 e^{-2C_x}$.

(b) If $\sigma_{\min}(\mathbf{B}(t)) \geq c_z$ *on* $[0,\varepsilon]$ *only, then* $\lambda_{\min}(\mathbf{G}) \geq \varepsilon\, c_z^2 e^{-2C_x}$.

In both cases $\sigma_{\min}(\mathbf{S}_1)^2 \geq \lambda_{\min}(\mathbf{G}) - \mathcal{V}$ *by Prop. 4. Case (b) is the specific regime of* $\mathcal{L}^{\ell}_{J_z,\mathrm{early}}$*. The upper-bound on* $\mathbf{A}$ *is needed on the whole interval as it enters through the transported factor* $\Psi(1,s)$*.*

The two terms of (15) have a different nature. The pointwise conditions on $\mathbf{B}$ and $\mathbf{A}$ considered above control $\lambda_{\min}(\mathbf{G})$ but they do not control the defect $\mathcal{V}$. This is why we design $\mathcal{L}_{\mathrm{sep}}$ to act after integration.

E.1 Proofs

Fix a unit $\mathbf{v}$ and set $\bar{\mathbf{m}} = \mathbf{S}_1\mathbf{v} = \int_0^1 \mathbf{M}(s)\mathbf{v}\,ds$.

$$\|\mathbf{M}(s)\mathbf{v}\|_2^2 = \|\mathbf{M}(s)\mathbf{v} - \bar{\mathbf{m}} + \bar{\mathbf{m}}\|_2^2 = \|\mathbf{M}(s)\mathbf{v} - \bar{\mathbf{m}}\|_2^2 + 2\langle \mathbf{M}(s)\mathbf{v} - \bar{\mathbf{m}}, \bar{\mathbf{m}}\rangle + \|\bar{\mathbf{m}}\|_2^2.$$

After integration the cross term disappears since we have $\int_0^1 (\mathbf{M}(s)\mathbf{v} - \bar{\mathbf{m}})\,ds = 0$. Therefore,

$$\|\mathbf{S}_1\mathbf{v}\|_2^2 = \mathbf{v}^\top\mathbf{G}\mathbf{v} - \int_0^1 \|\mathbf{M}(s)\mathbf{v} - \mathbf{S}_1\mathbf{v}\|_2^2\,ds \geq \lambda_{\min}(\mathbf{G}) - \mathcal{V}.$$

which is the identity (4). Then, taking the infimum over unit $\mathbf{v}$ gives us (15). For the corollary, the transition-matrix estimate of App. B, applied in reverse time, gives us

$$\sigma_{\min}(\Psi(1,s)) \geq \exp\left(-\int_s^1 \|\mathbf{A}(\tau)\|_2\,d\tau\right) \geq e^{-C_x}.$$

Hence, for every unit $\mathbf{v}$,

$$\|\mathbf{M}(s)\mathbf{v}\|_2 \geq \sigma_{\min}(\Psi(1,s))\,\sigma_{\min}(\mathbf{B}(s)) \geq c_z e^{-C_x}.$$

We then prove (a) by applying the squaring function and then integrating which gives $\mathbf{v}^\top\mathbf{G}\mathbf{v} \geq c_z^2 e^{-2C_x}$ for every unit $\mathbf{v}$. And to prove (b), we have $\sigma_{\min}(\mathbf{B}(s)) \geq c_z$ that is only assumed on $[0,\varepsilon]$, but the integrand $\|\mathbf{M}(s)\mathbf{v}\|_2^2$ is non-negative, so $\mathbf{v}^\top\mathbf{G}\mathbf{v} \geq \int_0^\varepsilon \|\mathbf{M}(s)\mathbf{v}\|_2^2\,ds \geq \varepsilon\, c_z^2 e^{-2C_x}$, and we conclude the same as before. □

F Architecture

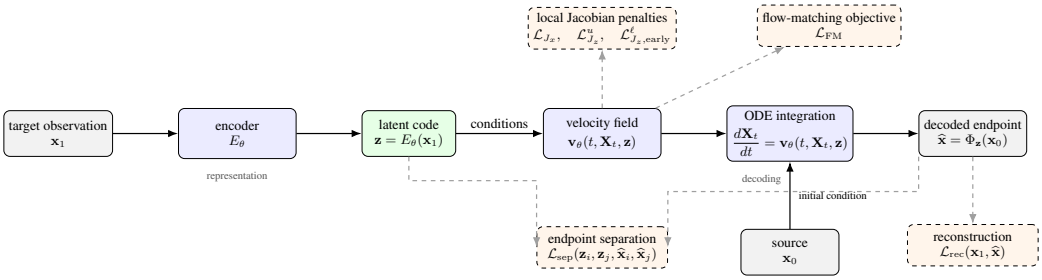


Figure 2: A target observation $\mathbf{x}_1$ is encoded into $\mathbf{z}$, which conditions the velocity field of a flow-matching decoder. Starting from a source $\mathbf{x}_0$ used as initial condition, ODE integration produces the decoded endpoint $\widehat{\mathbf{x}} = \Phi_{\mathbf{z}}(\mathbf{x}_0)$. Solid arrows: model computation. Dashed arrows: training objectives. The three local penalties act on the Jacobians of the velocity field and $\mathcal{L}_{\text{sep}}$ acts on the integrated decoder.

G Experimental details

G.1 Dimensions and protocol

Experiment	Input representation	Input dim.	Latent dim.
Synthetic gene expression	expression vector	100	2
Synthetic cell images	image $64 \times 64 \times 3$	12,288	12
Tabula Muris Senis	PCA projection of transcriptomes	256	16

Table 2: Spaces used in the three main experiments. For Tabula Muris Senis, 256 is the dimension of the PCA space given to the model.

The synthetic transcriptomic benchmark uses $5{,}000$ training and $2{,}000$ evaluation cells with 3 independent seeds. The rendered-image benchmark uses 5 paired seeds (identical initialization and data order across configurations). Tabula Muris Senis uses 5 paired seeds with two-fold cross-validation at the level of animal groups, so that no animal contributes to both training and evaluation. All comparisons between configurations share the same data and the same seeds.

G.2 Metrics

Reconstruction. $\text{MSE} = \frac{1}{N}\|\mathbf{x} - \widehat{\mathbf{x}}\|_2^2$, with N the observation dimension.

Factor recovery. For a continuous factor a known but never given to the model, we fit a linear predictor from the codes $\mathbf{z}$ and report $R_a^2 = 1 - \sum_i (a_i - \widehat{a}_i)^2 / \sum_i (a_i - \overline{a})^2$.

Empirical distortion. For test pairs, $r_{ij} = \|\widehat{\mathbf{x}}_i - \widehat{\mathbf{x}}_j\|_2 / \|\mathbf{z}_i - \mathbf{z}_j\|_2$. If a decoder preserves distance ratios, r_{ij} would be nearly constant. Large distortion means that some directions or regions are a lot more expanded than others. We summarize the dispersion by the robust ratio $\kappa^{95}_{\text{emp}} = r_{\text{p95}}/r_{\text{p05}}$, using percentiles to limit the effect of extreme pairs.

G.3 Synthetic data generation

Generative factors. Each observation is based on two continuous variables $(t, a) \sim \mathcal{U}([0,1]^2)$ with t the principal axis of variation and a the second one, deliberately made weaker. t and a are never given to the model.

Expression profiles. For $n_{\text{genes}} = 100$ genes, the rate of gene g is

$$\lambda_g(t,a) = \text{base}_g + A_g\, b_g(t)\big(1 + \rho_g(2a-1)\big) + C_g\, q_g(a), \qquad b_g(t) = \exp\Big(-\frac{(t-c_g)^2}{2w_g^2}\Big) \quad (16)$$

The t-dependent term produces activation waves centered at different points of the trajectory; $\rho_g \in [-0.5, 0.5]$ lets a modulate their amplitude without ever cancelling the effect of t; $q_g(a)$ is a smooth bump carried by a subset of genes that gives a its own direction of variation. The parameters $(\text{base}_g, A_g, c_g, w_g, \rho_g, C_g)$ are drawn once with a fixed seed and reused across every experiment, with $A_g \in [200, 400]$, so that all configurations see exactly the same observations.

This two-program construction deliberately avoids the alternative $\lambda_g(t,a) = \text{base}_g + aA_g b_g(t)$ that would actually map the whole edge $\{(t,0) : t \in [0,1]\}$ to a single expression profile. The synthetic manifold would then be non-injective which would make it impossible for us to distinguish a collapse observed in the latent from information which was already absent in the data.

Noise and preprocessing. Counts are drawn as $Y_g \sim \text{Poisson}(\lambda_g(t,a))$ and then transformed by $X_g = \log(1 + Y_g)$. A single global normalization is estimated on the training set and applied everywhere (no per-cell normalization is used as it would remove the amplitude differences a is meant to carry which is what we want to keep). We see that this benchmark has three distinct levels: (i) the known factors (t,a), (ii) noiseless profile $\lambda(t,a)$, (iii) and the noisy normalized observation. This allows us to attribute with good confidence a loss in the recovery of a to the representation rather than to the data.

Rendering to images. The same factors (t,a) drive a deterministic rendering layer producing $64 \times 64 \times 3$ images: t controls the dominant visual structure, a a continuous morphological attribute of smaller amplitude. A pixel-level linear probe gives $R^2_t = 1.000$ and $R^2_a = 0.847$.

G.4 Tabula Muris Senis: cohort and evaluation

The raw dataset [The Tabula Muris Consortium, 2020] contains 245,389 cells, 20,138 genes, 23 mice, 16 tissues, 123 cell types and six age groups. After preprocessing and cohort selection we keep 96,768 cells from 15 animal groups, 11 tissues and 5 age levels. The cohort is restricted to males because some age levels are represented by a single sex, which would confound sex and age. A context is a (tissue $\times$ cell type) pair. We keep contexts with at least 500 cells, at least 3 distinct age levels and at least 4 animal groups, giving 29 contexts over 8 tissues.

For each context, a linear age predictor is fitted on training animals and evaluated on held-out animals. Age is not involved in the training of the models. Cells are then projected on the fitted age direction, giving each cell a scalar coordinate along the learned age axis. The same diagnostics are computed in the model input space (the PCA projection), for the unregularized latent and the bi-Lipschitz regularized latent.

$\sigma_{\textbf{intra-bin}}$**.** The projected age axis is cut into bins and within each bin we measure the standard deviation of chronological ages and aggregate. A low value means cells close along the axis have similar ages.

Bins mixing far-apart ages. We divide the projected age axis into 20 quantile bins and report the fraction of bins containing cells from both the minimum and maximum chronological age represented in the corresponding held-out context.

Overlap 1 m / 30 m. The overlap of the projected distributions of 1- and 30-month cells, which measures the separation of the two chronological extremes without assuming a linear evolution between them.

Diagnostic (↓)	**Input space**	**Rec + FM**	**BFF**
$\sigma_{\text{intra-bin}}$	10.14	10.24	**7.08**
Bins mixing 1 m and 30 m	95%	95%	**60%**
Overlap 1 m / 30 m	57%	63%	**32%**

Table 3: Age-axis diagnostics on Tabula Muris Senis, aggregated over the 29 held-out contexts. Lower is better for all three. The values presented in this table are from the same seed as Fig. 1. The unregularized latent stays very close to the level of the input space and BFF improves all three.

H Scale invariance of the reported metrics

Could $\mathcal{L}_{\text{sep}}$ be satisfied by rescaling the encoder? m_z fixes an absolute scale between $\|\Delta z\|$ and $\|\Delta \hat{x}\|$, so a global rescaling $z \mapsto \alpha z$ can lower $\mathcal{L}_{\text{sep}}$ without reorganizing the latent. Both quantities we report are invariant under such a rescaling: κ^{95}_{emp} is a ratio of the ratios r_{ij}, each of which is divided by α, and R^2_a is obtained from a linear probe, hence is invariant under any invertible linear map of z. This justifies that the improvements of Table 1 are not the result of a scaling artifact.

I Extended limitations

Partial theoretical guarantees. The upper side has a direct justification through the bounds on the Jacobians of the velocity field. The lower side rests on mechanisms that are incomplete (the local guarantee near $t = 0$ (Prop. 2) + the sampled penalty on decoded endpoints). Their combination limits the crushing of latent directions but does not guarantee a uniform lower Lipschitz constant.

Insufficient ablation. The experiments in the main text compare the unregularized model with the full regularization only. This prevents us from analyzing and identifying separately the individual contributions of $\mathcal{L}_{J_x}$, $\mathcal{L}^u_{J_z}$, $\mathcal{L}^\ell_{J_z,\text{early}}$ and $\mathcal{L}_{\text{sep}}$. A term-by-term ablation would be a good future contribution to this work and could, for example, tell us if a simpler loss function formulation could retain the same benefits and if yes, invite us to understand why.

Choice of surrogate. $\mathcal{L}_{J_x}$ penalizes the Frobenius norm rather than the spectral norm directly. We do this as $\|J\|_2 \leq \|J\|_F$ for every matrix J. However, it is conservative since it also penalizes directions along which no expansion occurs. It is also a penalty on the average of $\|\partial_{\mathbf{x}} \mathbf{v}_\theta\|^2_F$ over sampled points. This does not enforce a uniform bound, so our penalization does not by itself deliver the constant C_x assumed in Prop. 1.

Unconstrained encoder. The geometric control of BFF acts on the decoder only. A natural next step is to add an encoder-side term, either a separation or neighbourhood-preservation constraint in the spirit of [Zhan et al., 2026], or a semantic regularizer aligning a latent direction with a known covariate, and to measure how much either adds on top of the decoder-side control.

J Additional results

J.1 Latent organization on the synthetic image benchmark

From the codes $\mathbf{z}$ we estimate by linear regression two directions: axis_a, which best predicts a from $\mathbf{z}$, and axis_t, which does the same for t. Each code is projected on both. If a is preserved as an axis of variation we expect the codes to be spread along axis_a rather than confined to a narrow region, and the value of a to vary in an ordered, continuous way along it.

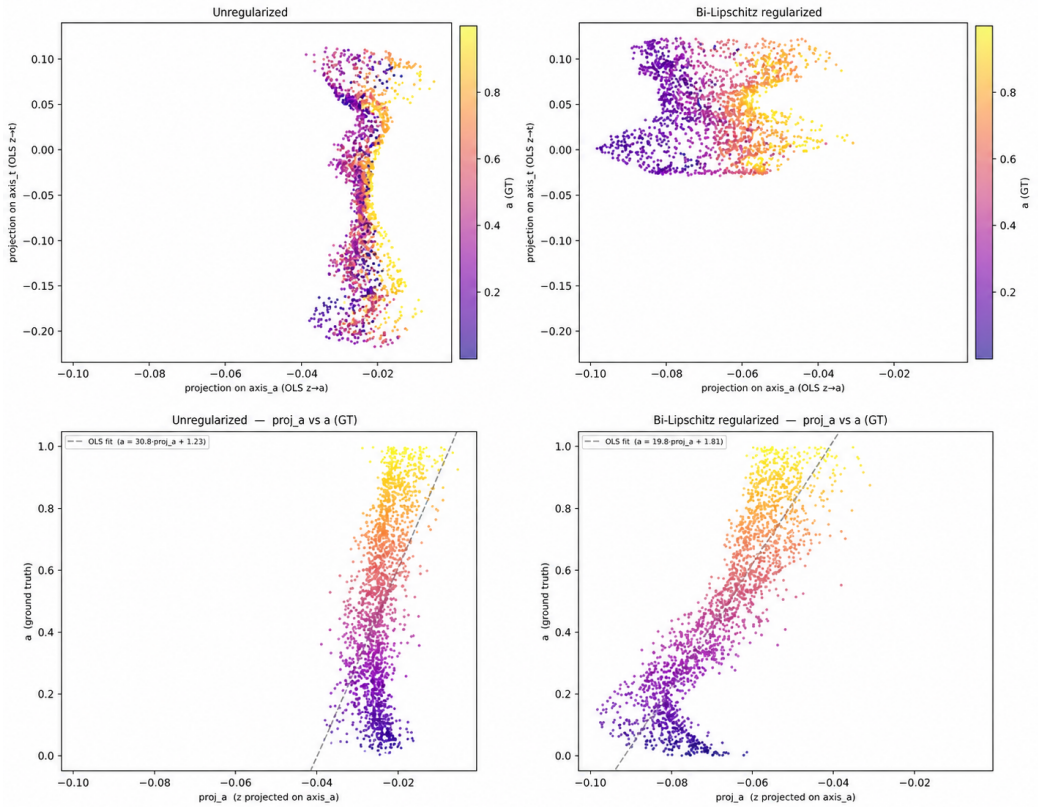


Figure 3: Latent organization of the sub-dominant factor a on the synthetic image benchmark. **Top:** test codes projected on the OLS directions associated with a (horizontal) and t (vertical), colored by the true value of a. **Bottom:** the projection along axis_a against the ground-truth factor. The unregularized model shows codes that are compressed along the a-direction and different values of a land in nearby latent regions. BFF shows codes that are more spread out and the colors order progressively. The seed shown is the one whose R^2_a under BFF is the median of the five runs, and it is the same seed used for the unregularized model.

J.2 Towards real cell imaging

The long-term goal is to apply the same principle to real cell images, where flow matching has a clearer structural advantage than in vector reconstruction problems. A first implementation on BBBC021 [Caie et al., 2010] was conducted but was not carried far enough to provide a meaningful quantitative result, and is therefore not reported here.